

KUSH MEHTA

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EDUCATION

Institute of Technology, Nirma University

Ahmedabad, India

B.Tech, Electronics & Communication Engineering — Minor: Data Science | **CGPA: 8.43 / 10**

Aug 2023 – May 2027

- Relevant Coursework: Embedded Systems, Computer Architecture, VLSI Design, Digital Signal Processing, Image Processing, Signals & Systems, Research Methodology & Seminar, Communication Systems, Scripting Languages
- Minor in Data Science — coursework in statistics, data analysis & visualization, machine learning, Analytics of IoT

EXPERIENCE

Electronics Lead — Team Arrow, Nirma University

Sep 2023 – Present

Nirma University Official Aeromodelling & UAV Club

Ahmedabad, India

- Led electronics design and integration for fixed-wing competition aircraft across SAE Aero Design West 2025-**5th Internationally(mission)**, SAEISS DDC 2024 & 2025-**AIR 2**, and Technoxian 2024-**4th Internationally**
- Designed and implemented autonomous systems, flight controller tuning (ArduPilot/Pixhawk), telemetry links, and onboard payload electronics
- Achieved multiple top-10 national and international podium finishes across 4 competitions
- Mentored junior members on avionics integration, autonomous systems, and embedded firmware

PROJECTS

[LINK](#)

Autonomous Tilt-Rotor VTOL Tricopter with AGV Payload

2025-2026

- Tilt-rotor tricopter (NACA 9325 fuselage, S1223 wings, 800mm span, 3.415 lbs); 2:1 thrust-to-weight, ~4 min hover
- Full mission: autonomous takeoff, cruise transition, target capture by AGV, return — validated end-to-end in SITL before hardware deployment

Secure GPS Location Tracker — ESP32 + AES-128 + MQTT

Mar 2026

- Built end-to-end IoT security pipeline: ESP32 reads GPS via TinyGPS++, serializes to JSON, encrypts with AES-128 (mbedTLS), publishes over MQTT; Python subscriber decrypts and auto-opens Google Maps
- Validated at Nirma University (23.13°N, 72.54°E); encrypted hex stream confirmed on HiveMQ broker

IoT Cryptography Benchmark — AES vs ASCON vs DES vs ECC

Jan 2026

- Benchmarked 4 cryptographic algorithms across 5,000 IoT JSON payloads; ASCON (loaded via ctypes DLL) achieved 78× speedup over AES-GCM with peak memory of 320 bits
- Used Python tracemalloc for memory profiling; results validated ASCON as optimal for resource-constrained edge devices

gem5 L1 Cache Sensitivity Analysis

Feb 2026

- Simulated X86 TimingSimpleCPU on gem5 v25.1 with matrix multiply and AES workloads; 4kB→64kB cache growth reduced miss rate from 5.06% to 0.04%, IPC +11.9%
- 8-way set-associativity reduced miss rate from 4.74% to 0.34% with IPC gain of +10.4%; concluded highly-associative modest caches optimal for UAV/IoT applications

Virtual Try-On Sunglasses — Real-Time AR (OpenCV)

Nov 2025

- Classical CV pipeline: CLAHE preprocessing → Haar Cascade face/eye detection → geometric pose estimation → alpha blending; EMA smoothing ($\alpha=0.2$) eliminates jitter
- HSV hue-shifting (mod 180) enables live color customization without affecting brightness or transparency

TECHNICAL SKILLS

Languages: Python, C/C++, Verilog HDL, Bash, MATLAB, SQL, Assembly Language

Embedded & IoT: ESP32, Raspberry Pi, Arduino, FPGA, Keil, ArduPilot, UART/I2C, MQTT, Node-Red, PyCrodome

Data Science: Pandas, NumPy, Scikit-Learn, Matplotlib, Tableau, KNIME, OpenCV

Tools: Git/GitHub, Linux, SolidWorks, Fusion 360, gem5, Solidworks, Simulink, Quartus II, Modelsim

RESEARCH

Simulation & Comparative Analysis of Lightweight Cryptography for Embedded Systems 2025–2026

Research Methodology & Seminar — Nirma University

In Progress

- Systematic comparison of AES, DES, ECC, and ASCON on throughput, memory footprint, and energy efficiency across constrained hardware profiles; ASCON identified as optimal NIST lightweight standard